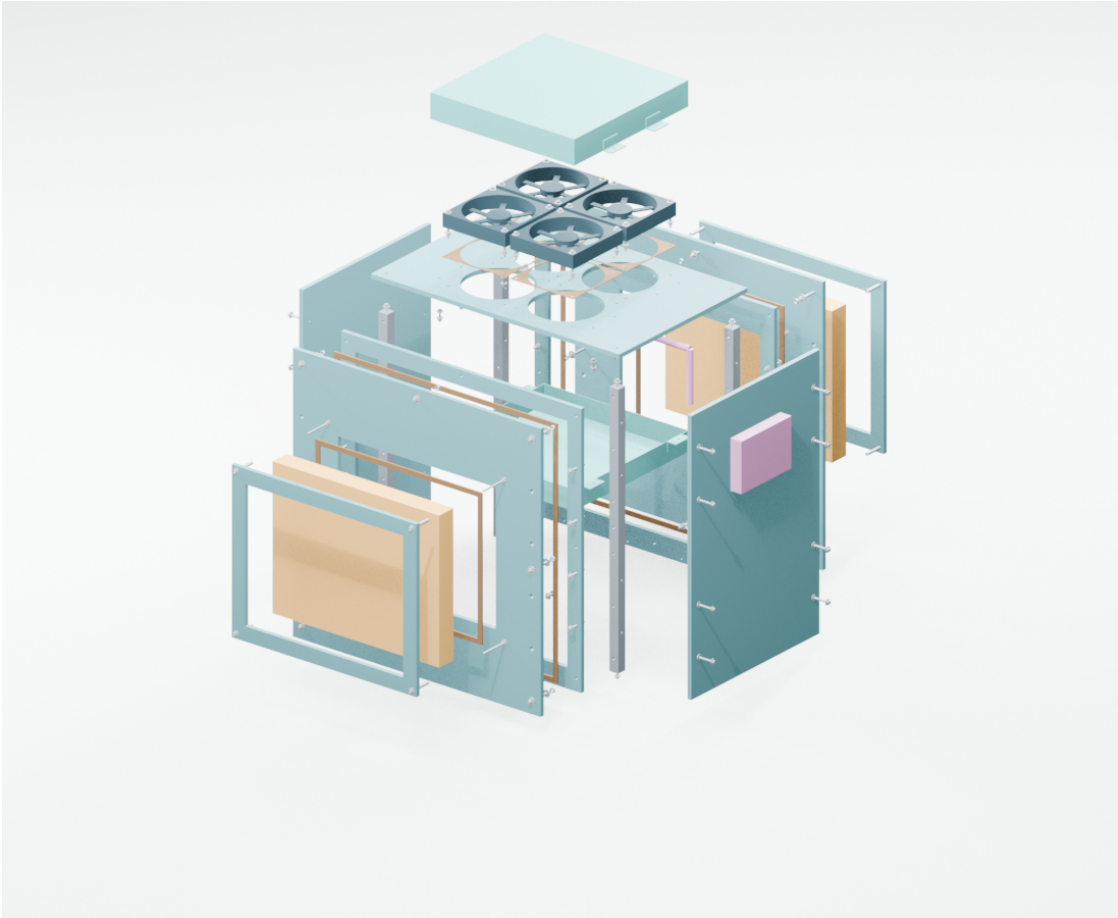


AQI Tower Engineering Handoff

Current D01 R03M integration with IH02 evidence



NOT FOR FABRICATION OR ENERGIZATION. Authoritative manufacturing geometry remains the FreeCAD R03M study; Blender and browser meshes are display derivatives.

The following review pages lead into the unchanged 41-page IH02 engineering reference. Return named decisions with evidence in TECHNICAL_REVIEW.xlsx. Do not mix legacy large-tower mains/HEPA assumptions with this 12 V indoor demonstrator.

Current pressure and geometry

Quantity	Current screen or evidence
Total flow	300 m3/h assumed, not measured
Parallel branches	4 fans at 75 m3/h each; 2 filters at 150 m3/h each
OEM fan head	27.63 Pa at stated curve condition
Nonfilter loss	13.31 Pa under K1 reducer scenario
Remaining filter head	14.32 Pa per parallel filter branch
CAD inventory	493 source objects; tessellated display derived from saved R03M
Display bounding envelope	595 x 504 x 631 mm including reservations and projecting hardware; not a fabrication specification

The two parallel filters do not add their pressure drops in series. Clean/loaded filter curves, installed/static fan curve compatibility, loss coefficients and branch equality remain unresolved. This screen is not an operating-point prediction.

Colour and animation interpretation

Teal identifies cabinet/guards, amber filters, dark blue fan envelopes, purple reserved control space. Motion shows part separation or hand-authored air paths, not velocity, turbulence, deposition or removal. Guard faces remain solid tessellations, not the proposed perforation geometry.

Review responsibilities and construction path

Discipline	Closure required
Mechanical	Guard strength/reach, seams and attachments; cabinet joints and stability; real filter tolerance, gasket compression, safe service access and whole mass/CG.
Electrical	Hazard allocation, independent protective functions, actual DC switching/isolation/protection, wire/connector ratings, cable restraints and enclosure. No guessed terminal assignments.
OEM-dependent	Exact fan/extension connector and clamping dimensions; delivered filter/frame properties. Sent enquiries are pending.
Commissioning and test	Confirm premises and targets; review assembly, isolation/restart and abnormal behaviour before collecting measured performance.

Critical path

Exact component evidence and accepted targets -> mechanical/electrical review -> revision-specific fabrication release -> supplier delivery and fabrication -> assembly and qualified commissioning -> airflow/pressure and controlled particle tests. These dependencies are not completion dates.

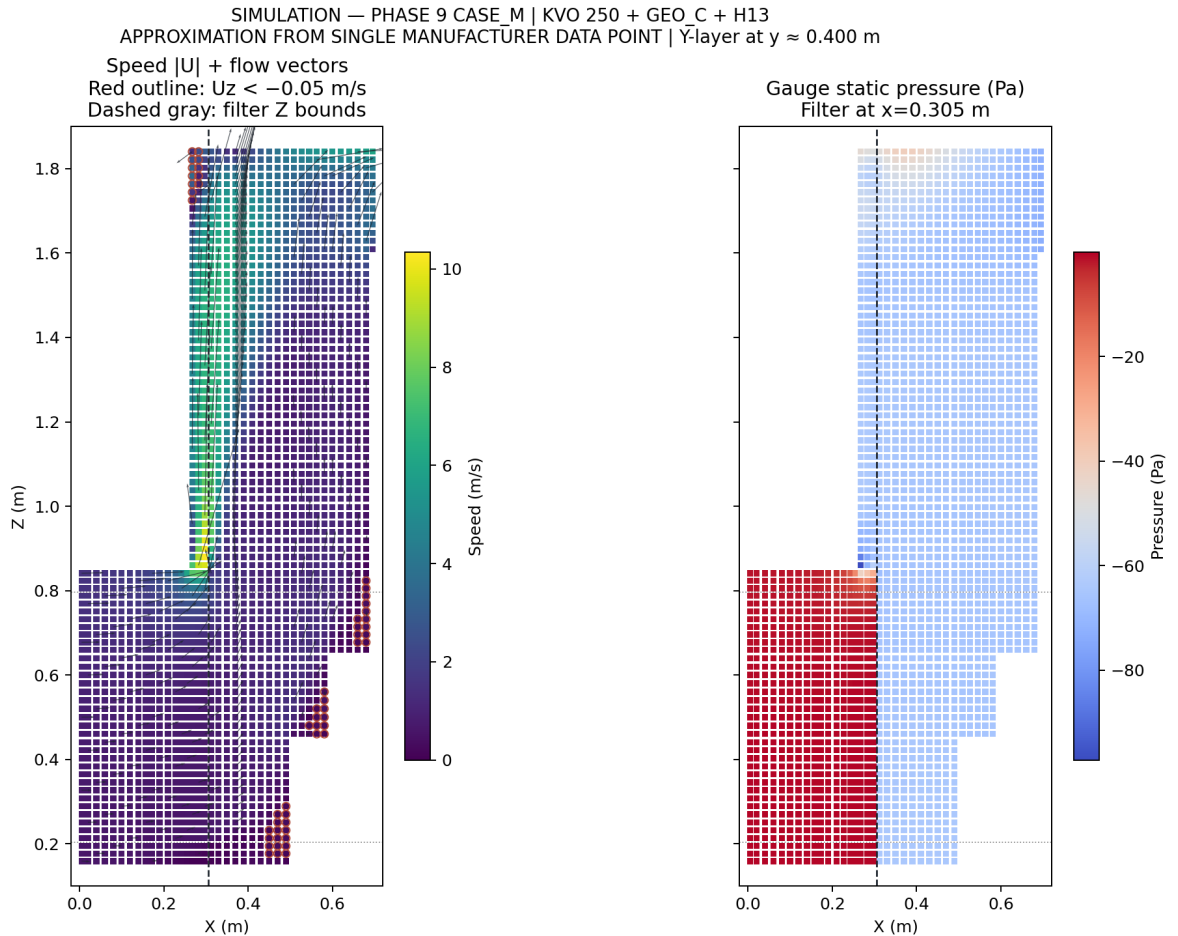
Outdoor boundary

Current fan temperature limits, enclosure and materials do not establish suitability for Indian outdoor heat or public misuse. Weather protection, tamper-resistant service access, anchoring and wind-dependent performance require a separate qualified design and approved site.

Files to use

Native CAD, STEP, 15 DXFs, current tools, measurement templates and source registers are in engineering/. TECHNICAL_REVIEW.xlsx combines parts, release, pressure and hardware sheets. The Blender file is for presentation only.

Historical CFD evidence



Existing Phase 9 CASE_M plot: KVO 250 + GEO_C + H13 approximation from a single manufacturer data point. Colours show calculated speed and gauge pressure for that historical case, not R03M performance.

GEO_C is an earlier air-path geometry trial. It is not the current manufacturing assembly. Historical CFD is partial/nonconverged; the later numerical bypass seal is not a proven physical seal. No new R03M CFD has been run for this delivery.

For the current prototype, use the pressure screen and obtain measured filter/installed-flow evidence. The following IH02 appendices are preserved unchanged.